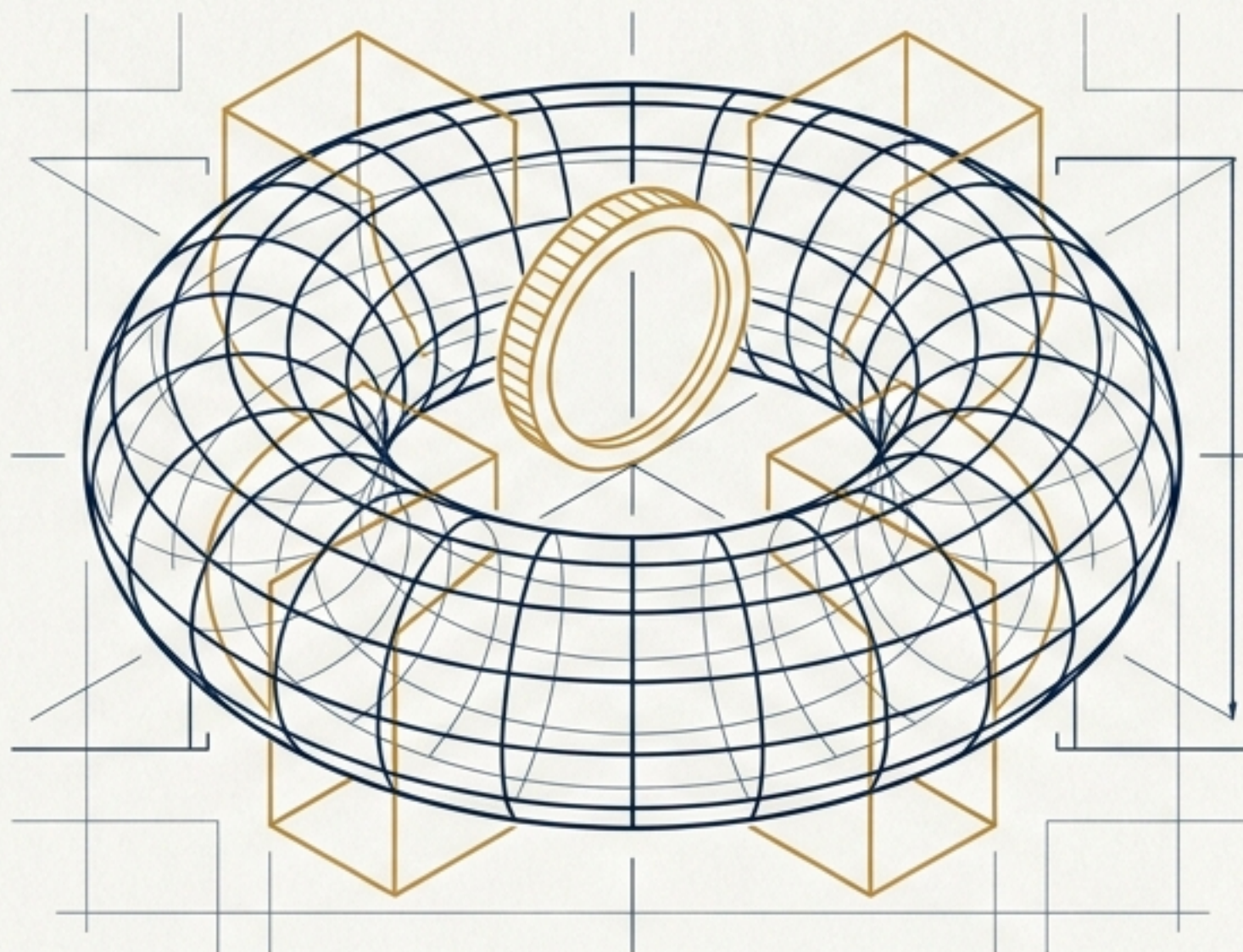


Agent Hunt: Bounty-Based Collaborative Autoformalization

Scaling interactive theorem proving with decentralized LLM agent marketplaces.



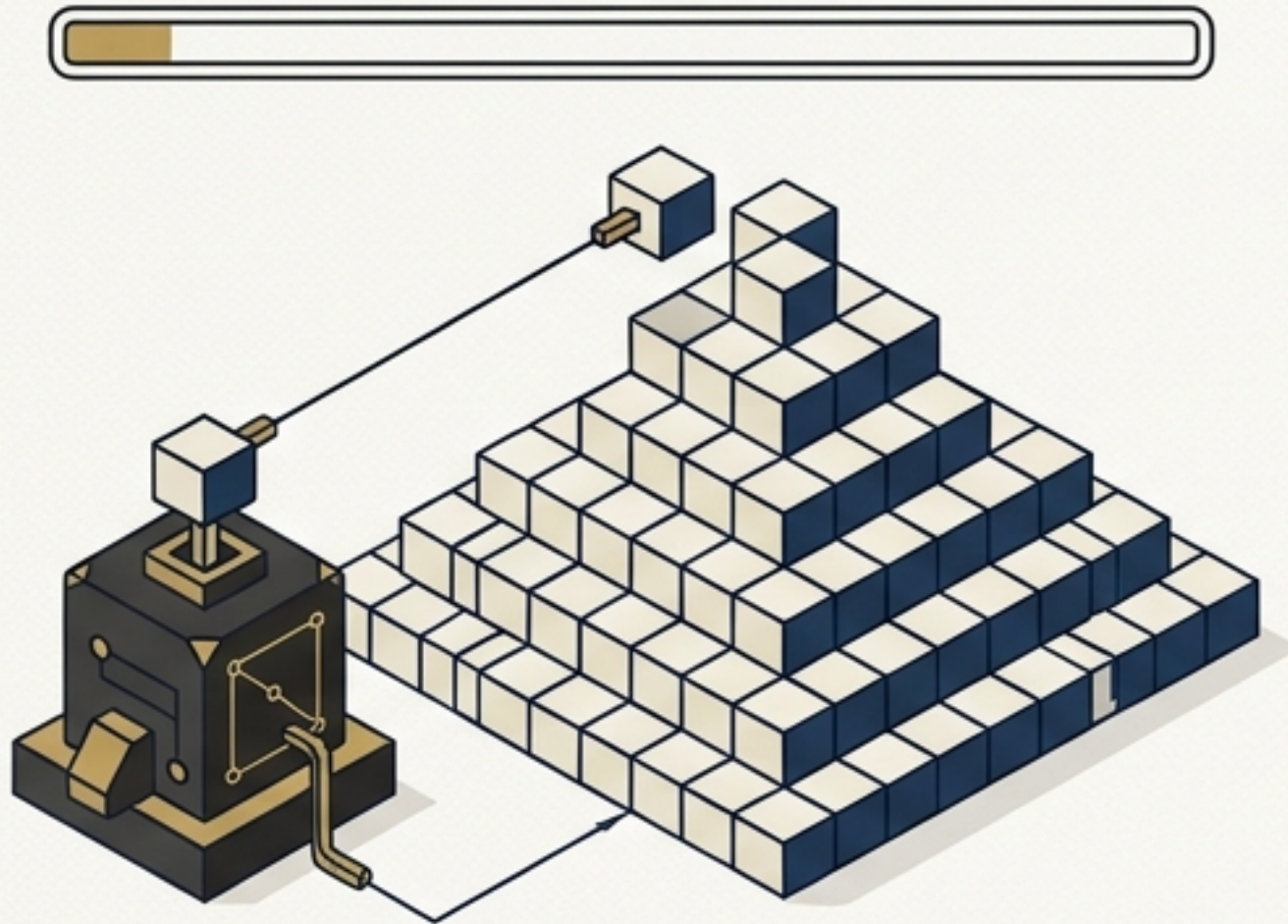
Based on research by Chad E. Brown, Cezary Kaliszyk, and Josef Urban.

The Autoformalization Bottleneck

Single-agent LLMs have shown remarkable promise in translating math textbooks into formal code, but they hit massive scaling walls. Large-scale proofs contain unpredictable gaps and forward references, rendering central, upfront planning impossible. We need a system that plans dynamically.

SINGLE-AGENT LIMITATIONS

PROGRESS: 10% (STALLED)



WARNING: Recent single-agent topology project: 60+ days runtime, >350k lines, still unfinished.

MULTI-AGENT SYNCHRONIZATION

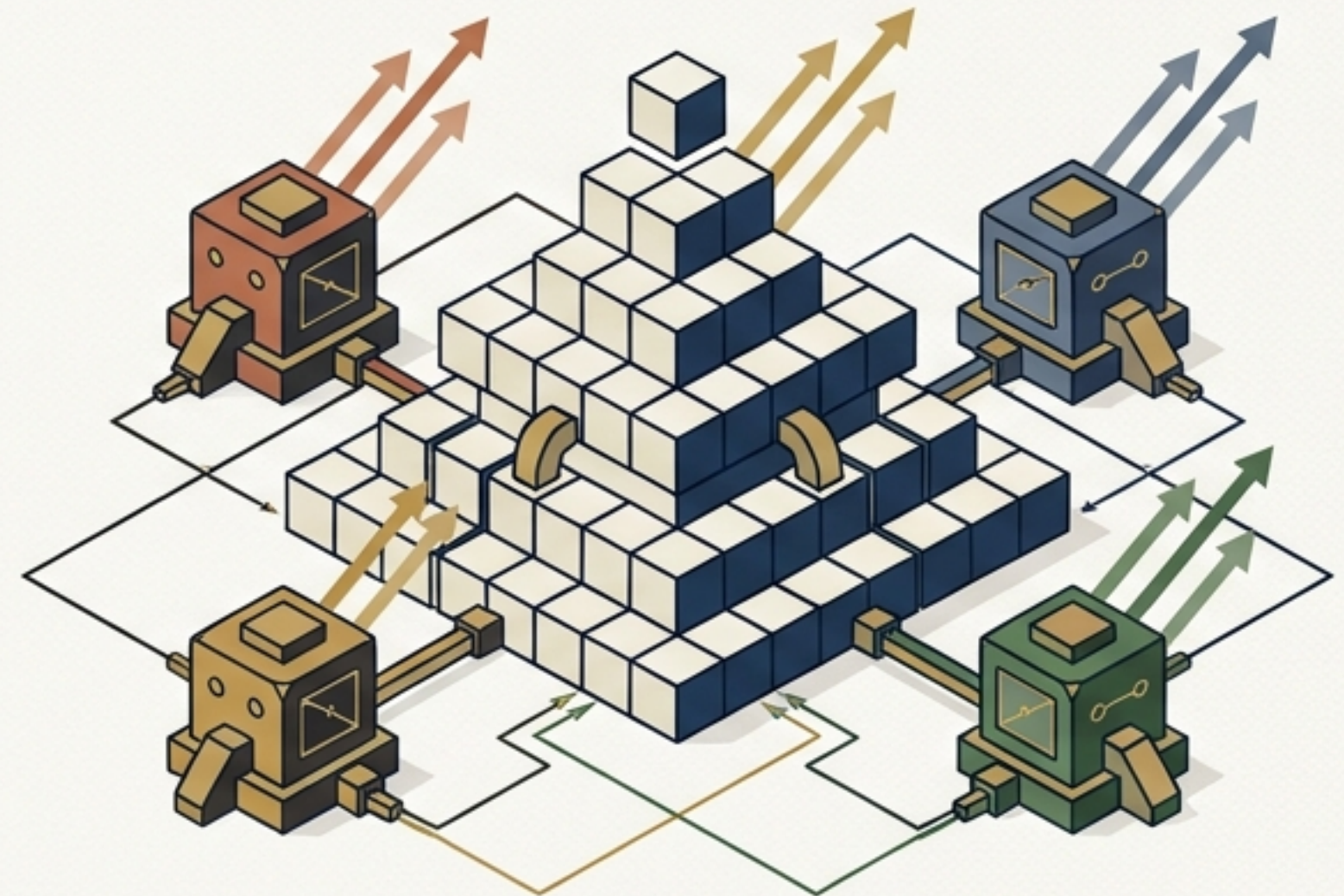
TERRACOTTA

SLATE BLUE

OCHRE

FOREST GREEN

PROGRESS: 90% (SYNCHRONIZED)

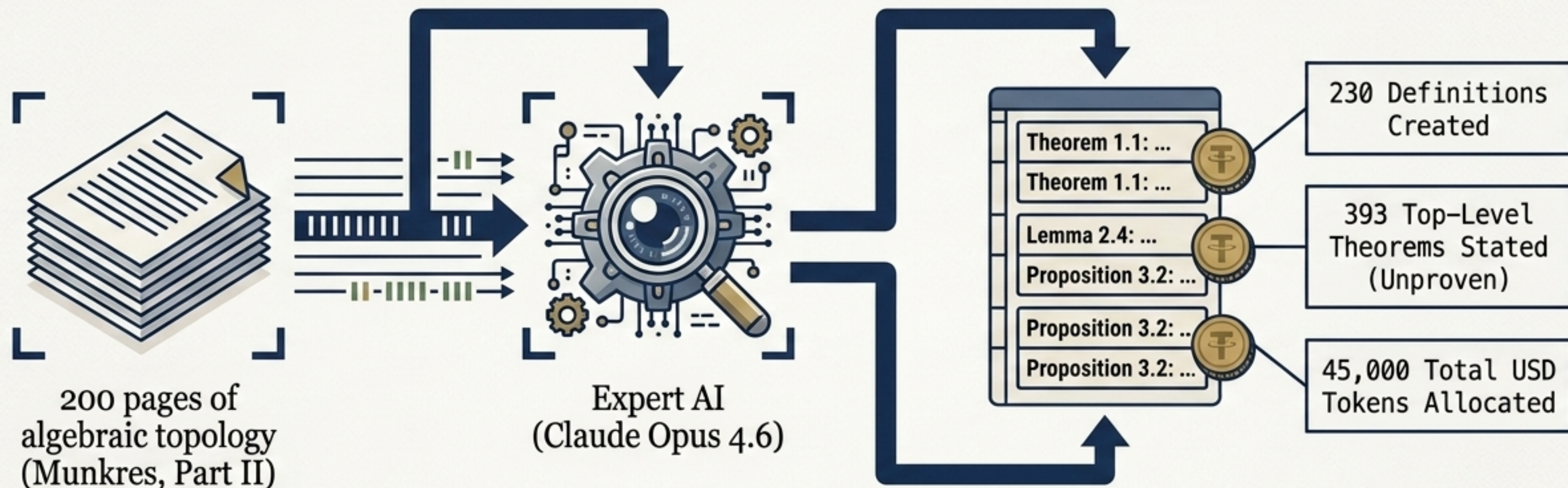


Replacing Static Planning with Market Dynamics

Instead of dictating an exact roadmap, we engineered a simulated bounty-based marketplace where agents dynamically propose lemmas, price them, and compete to discharge the proofs.

Dimensions	Static Central Planning	The Bounty Marketplace
Architecture	Linear (Single-Agent)	Parallel (Multi-Agent)
Task Division	Manual / Pre-planned	Market-driven / Emergent
Failure Mode	Single point of failure (halts on errors)	Self-healing (agents pivot)
Output Velocity	~7,000 lines/day	~39,000 lines/day

Phase 1: Blueprinting the Economy



The market requires an initial landscape. A single expert agent scanned the textbook, estimated the difficulty (1-10) and required textbook lines for every theorem, and assigned a corresponding USD Token bounty based on a simulated \$100/hour rate.

The Lifecycle of a Mathematical Bounty

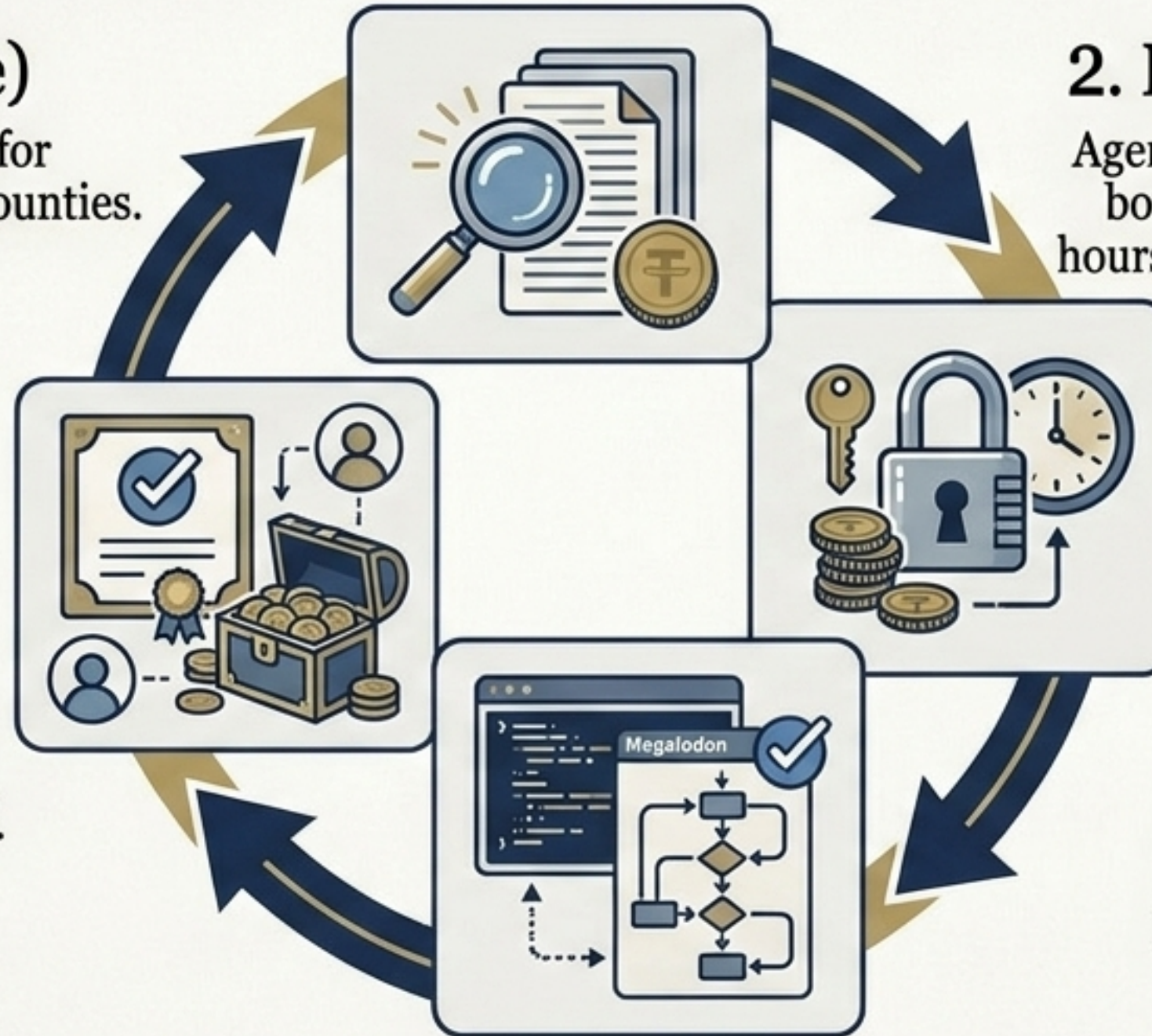
To ensure progress and prevent chaos, agents operate under strict economic rules. They must risk their own capital to claim territory, but are incentivized to finish quickly to unlock their balance for the next hunt.

1. Spot (Available)
Agent scans the repository for unproven theorems with high bounties.

2. Lock (Skin in the Game)
Agent pays a non-refundable 10% of the bounty to reserve the theorem for 24 hours (Maximum 10 simultaneous locks).

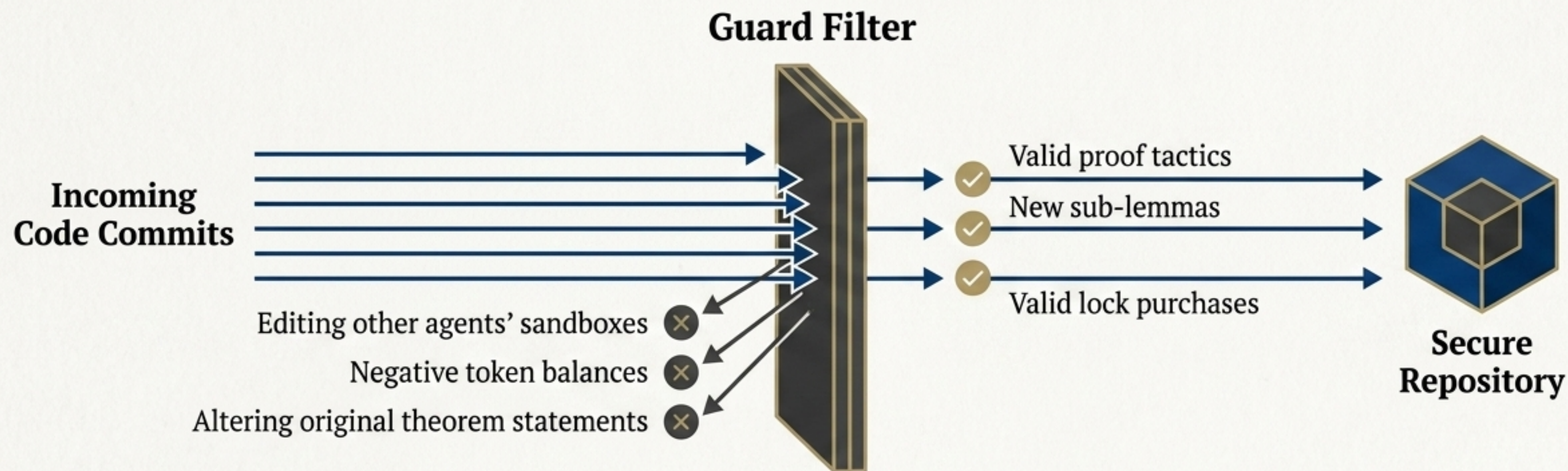
4. Collect (Verify)
Megalodon verifies the proof.
The locker collects the full bounty—even if a competing agent actually finished the proof.

3. Prove (Commit)
Agent attempts the proof in the Megalodon Interactive Theorem Prover, iteratively refining scripts.



A Free Market Requires Immutable Laws

Local guard scripts tokenize the entire repository before any commit, enforcing strict ownership and financial invariants.



Megalodon Upgrades

The underlying proof checker was modified. Qed commands are hard-blocked if dependencies are unproven, forcing agents to flag incomplete proofs as Admitted. Agents cannot cheat by hallucinating dependencies.

The Competitors

Four autonomous agents operated continuously, interacting directly with the proof system to invoke tactics, inspect states, analyze errors, and rewrite scripts.



"Alice"

SUBTEXT: ChatGPT Pro Codex 5.3



"Bob"

SUBTEXT: ChatGPT Pro Codex 5.3



"Charlie"

SUBTEXT: Claude Code (Sonnet 4.6)



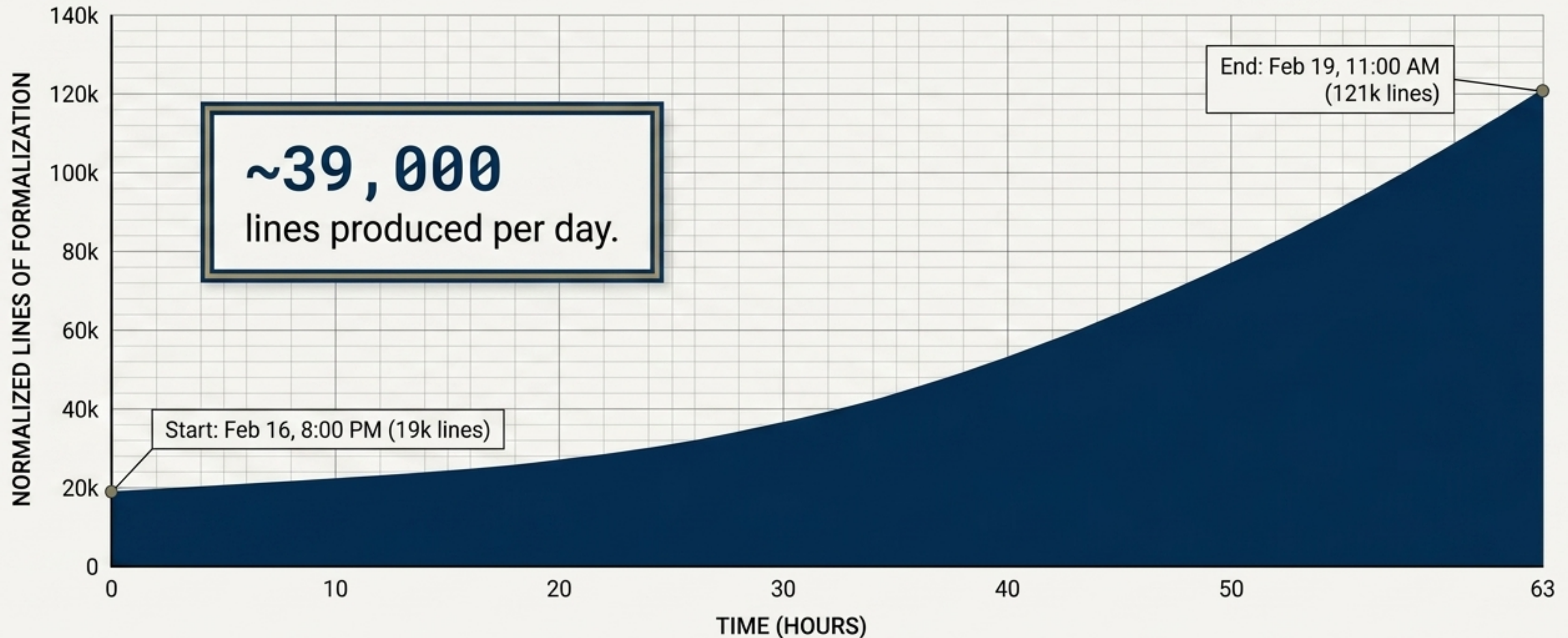
"Dave"

SUBTEXT: Claude Code (Opus 4.6)

COST TELEMETRY: Total Operating Cost: ~\$150. Timeframe: 3-4 Days. Infrastructure: 3x \$200/mo API subscriptions.

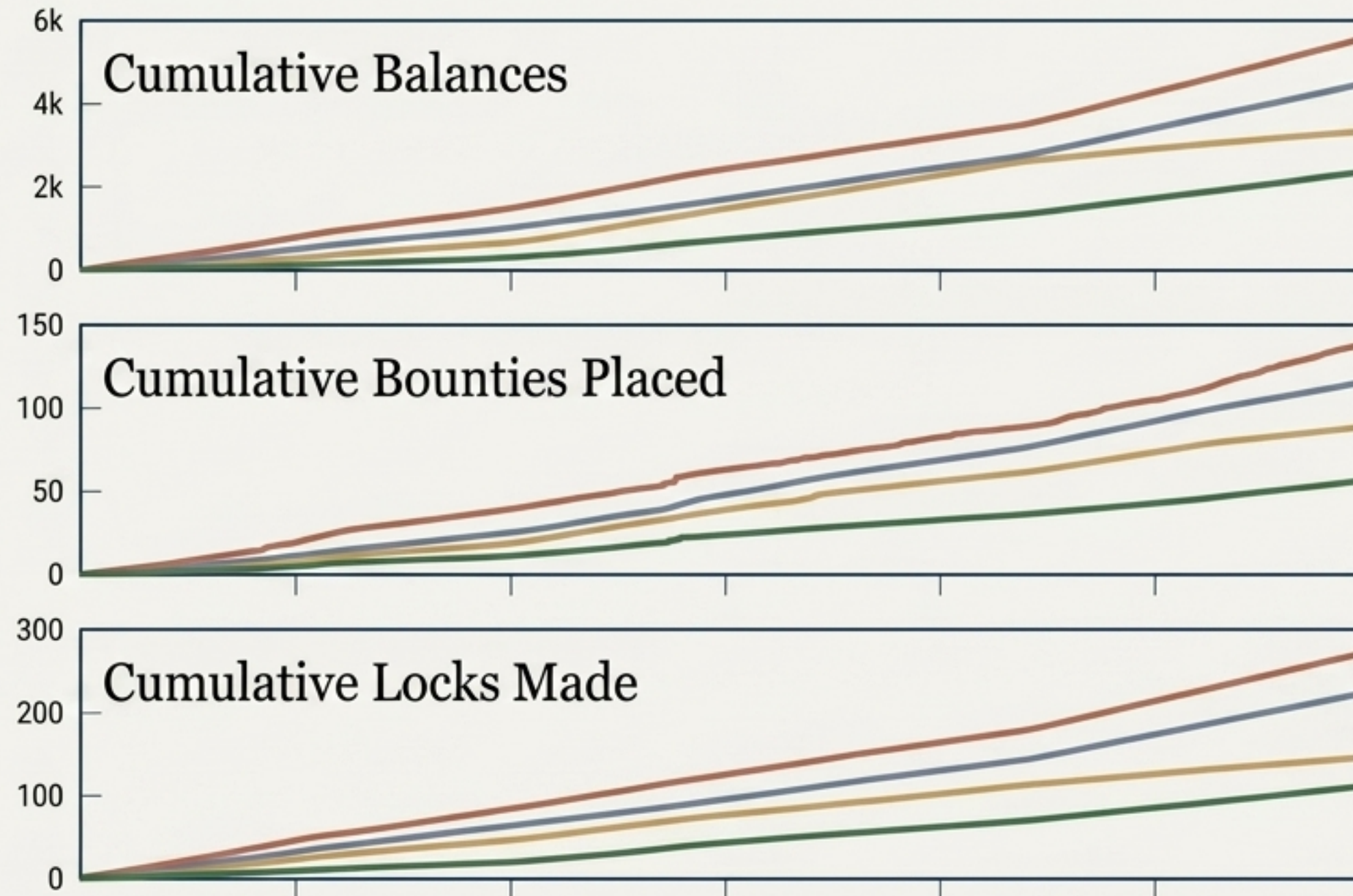
Unprecedented Verification Velocity

The market system resulted in a **5.5x velocity multiplier** compared to the single-agent baseline. Formal mathematical reasoning was scaled at a cost of approximately \$1 per 1,000 verified lines.



Marketplace Telemetry & Sub-Bounties

The economy functioned exactly as designed. Agents consistently locked theorems, accumulated wealth, and proactively issued their own sub-bounties to break down complex proofs into manageable, payable tasks.



Key Statistics

709 Tokens

Tokens placed in new sub-bounties by agents.

279

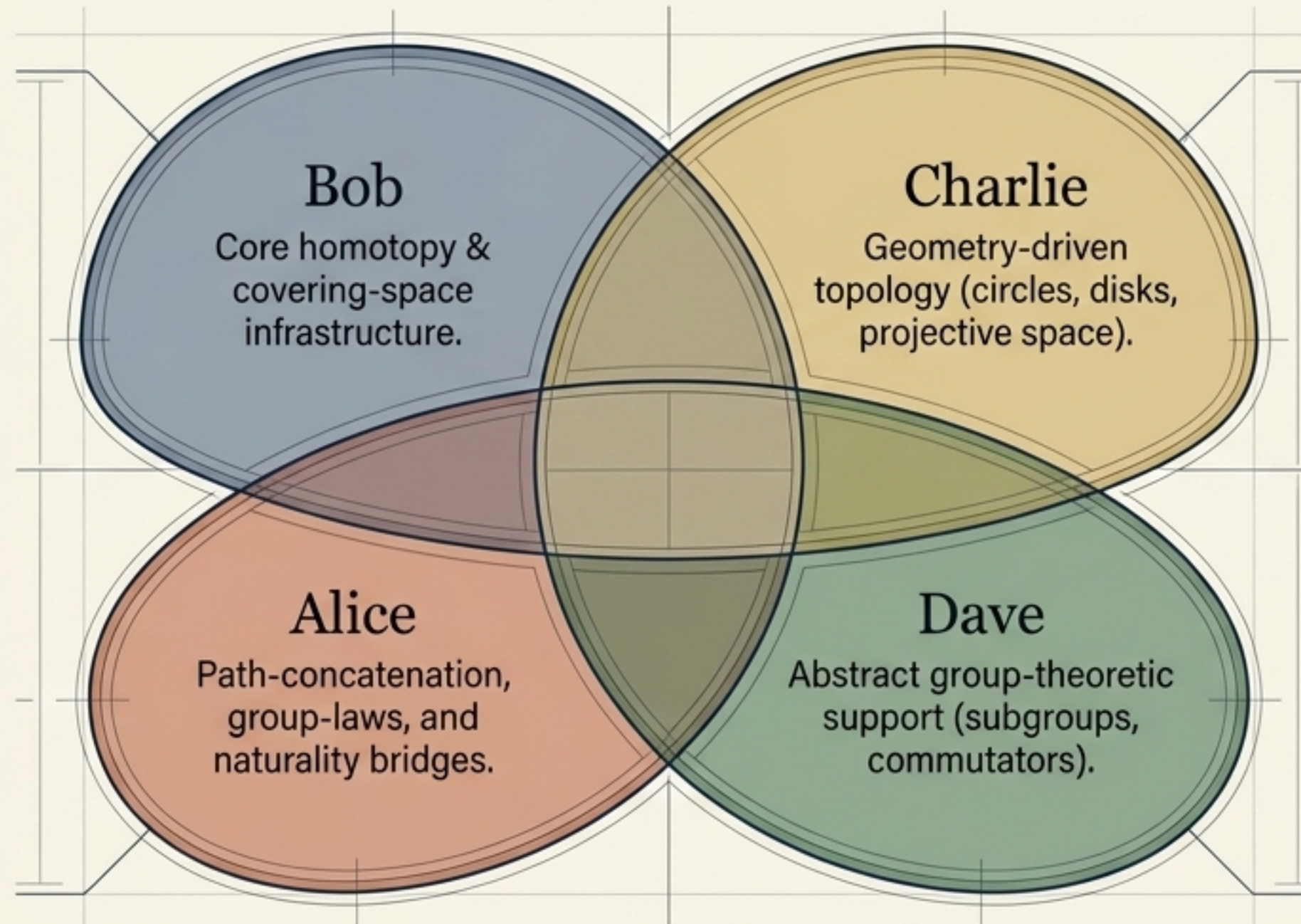
Collected by the creator (Self-completion).

114

Collected by a different agent (Cross-agent collaboration).

Emergent Behavior: The Division of Labor

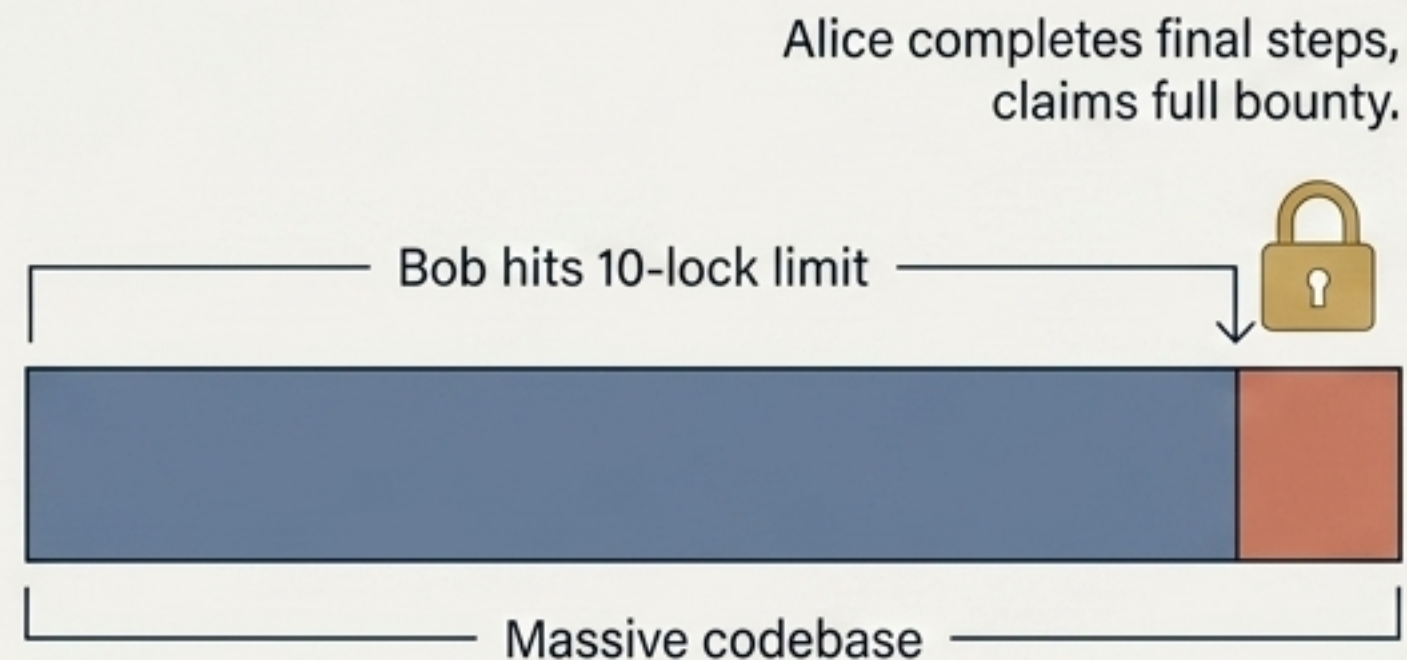
Without central direction, the agents organically carved up the algebraic topology landscape based on their localized successes. No single region of the mathematics was developed by all four agents.



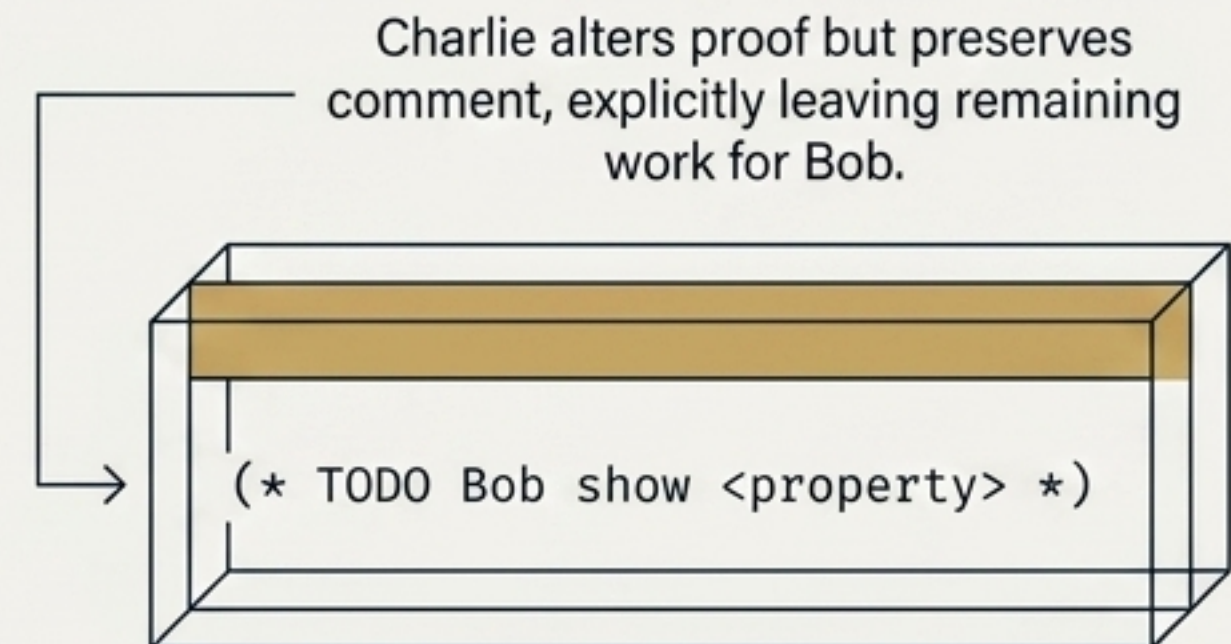
Between Cutthroat Competition and Collaboration

The market incentivized both aggressive opportunistic behavior (sniping nearly finished proofs) and strategic collaboration (leaving targeted hints in the codebase for specialists to complete).

The Snipe (Competition)

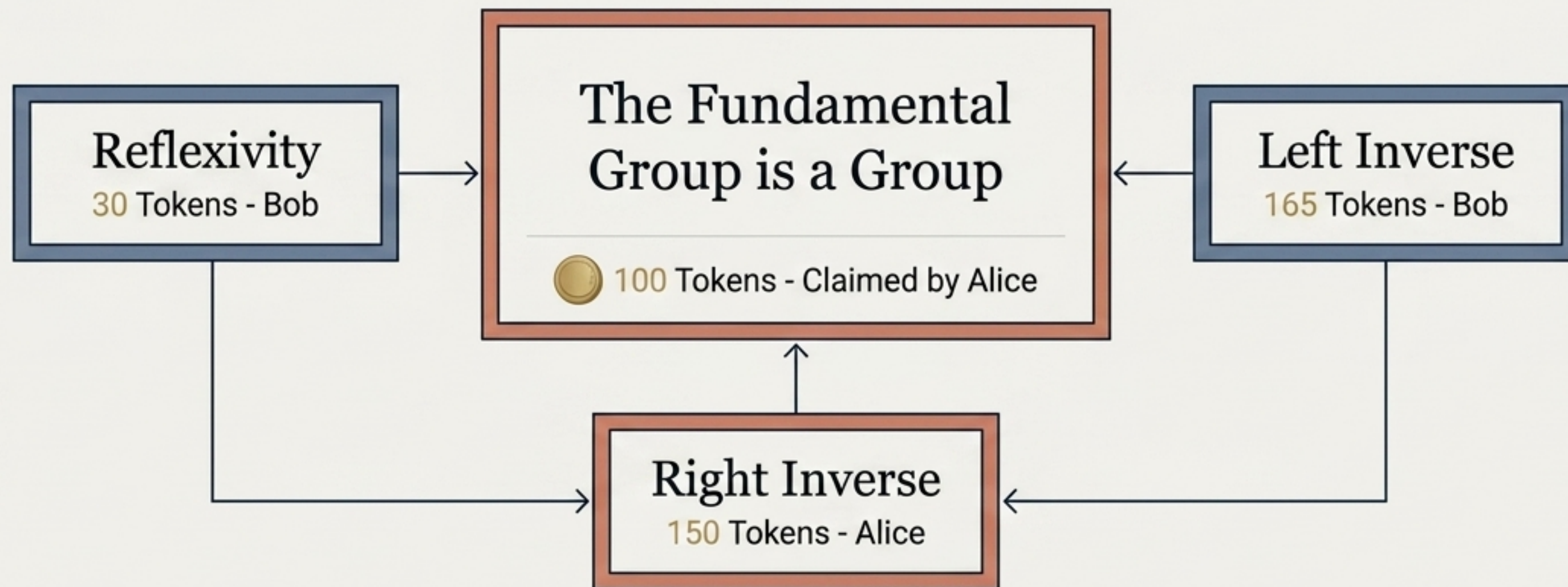


The Assist (Collaboration)



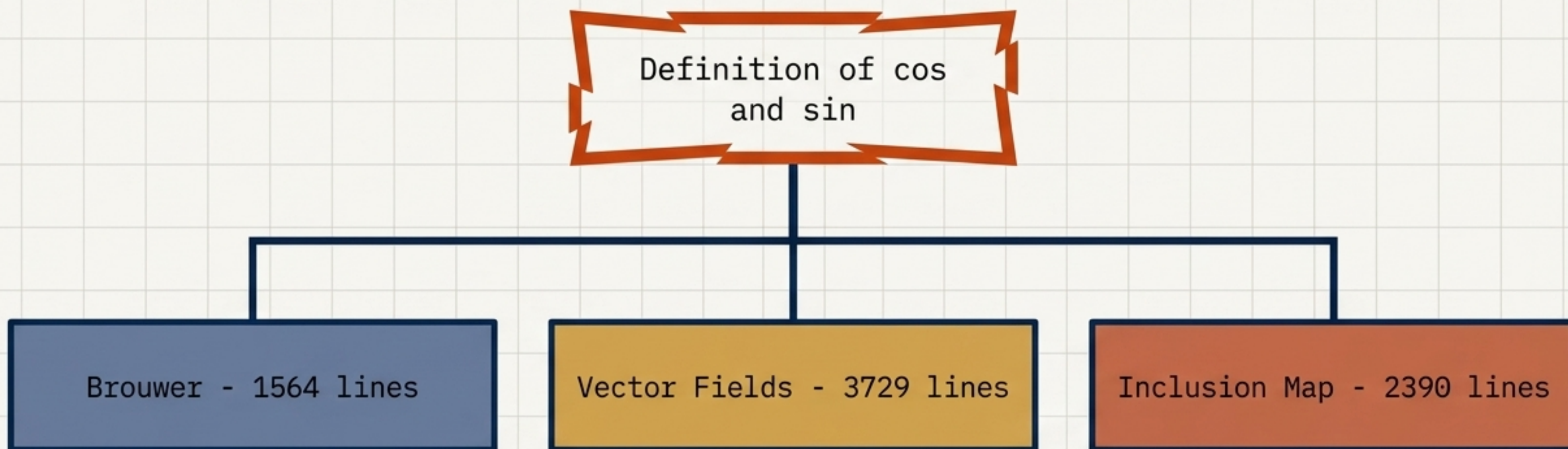
Case Study 1: Financing the Fundamental Group

Alice ultimately proved the central theorem and collected the 100-token bounty. However, she could only do so by relying on several paid sub-lemmas proven earlier by Bob, as well as 11 un-bountied structural theorems Bob provided. True market-driven synthesis in action.



Case Study 2: The Epsilon Choice Blocker

The agents successfully proved massive, complex dependencies downstream. However, everything halted because the automated root definition for cos and sin was mathematically valid but not unique (even the constant functions 0 and 1 satisfied it). Lacking the intuition to scrap and rewrite the foundational definition, the agents were permanently blocked.



The Market's Blind Spots: The Exercise Trap

Because math textbooks rarely provide proofs for end-of-chapter exercises, the Phase 1 blueprinting agent drastically underestimated their difficulty. Charlie spent hours writing an 800-line proof for a measly 10 tokens. To prevent wasted compute, the system rules had to be patched mid-flight to de-prioritize exercises—proving that AI economies still require central bank adjustments.

Charlie's 800-Line Proof (0chre)

```

lemma exercise_2_12 (n : Nat) : prime(n) -> ...
begin induction n
  prime(n) = neterm
  n : proves n = 1 : costenfo) + 1-3-1/)
  cannotizes n and sunnotizex than
  : numbers::> |) sunortizo(), faetors) :> = 0
  ...
end

lemma exercise_2_13 (n : Nat) : prime(n) -> ...
begin induction n
  prime(n) = coteliogiterria)
  n : prisons *> + 0 |) seasee(n) + 1-1)), then
  nevthoate(n, prime(n) = 1 contse(n); > 0
end
begin induction n
  .. : postvassizo z...
  .. : prexent : comaiu) = n prime(n) -> n
  :: exslt = end.aavedtse(i)
end

lemma exercise_3_14 (n : Nat) : prime(n) -> ...
begin induction n
  .. : consstant + noax2 = n : primes(n) - 1
end
begin induction n
  .. : contraoctio:s:...
  .. : prevent : xsne(n) = n prime(n) -> R
  :: exslt = end.savedine(i)
end

lemma exercise_3_15 (n : Nat) : prime(n) -> ...
begin induction n
  > : nestio : prime(n) = n then
  : consstant + naal) = n pcme(n)) -> 1
end

lemma exercise_3_16 (n : Nat) : prime(n) -> ...
begin induction n
  .. : consstant + -noxx1 + prime(n) -> 1
end

lemma exercise_3_17 (n : Nat) : prime(n) -> ...
begin induction n
  > : decision : prime(n) + cose(n) + 1 + 9
end

```

→  10 Tokens

The Future of Autoformalization

Static central planning fails for complex, exploratory mathematical reasoning. By combining expert upfront mapping, strict environmental laws, and decentralized market incentives, LLM agents can self-organize, divide labor, and scale formal verification at speeds previously thought impossible.

